

MECE Breakdown and Insights Document

Sakila DVD Rental Store Analysis



Submitted By

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Project Title

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Contents

1	Introduction	5
2	Entity Relationship Diagram (ERD)	5
2.1	Key Relationships	6
2.2	Importance of the Data Model	7
3	Customer Analysis	7
3.1	Active vs. Inactive Customers	7
3.2	Top Customers by Revenue	7
3.3	Detailed Customer Information	8
3.4	Rentals vs. Revenue Analysis	8
3.5	Geographic Distribution of Customers	8
3.6	Key Performance Indicators (KPIs)	8
4	Revenue Analysis	9
4.1	Total Revenue	9
4.2	Monthly Revenue Trend	9
4.3	Average Payment	9
4.4	Top Revenue-Generating City	10
4.5	Revenue by Film Category	10
4.6	Rental Revenue by Country	10
4.7	Revenue Distribution by Film Rating	10
4.8	Key Performance Indicators (KPIs)	10
5	Rental Analysis	11
5.1	Total Rentals	11
5.2	Returned Rentals	11
5.3	Active Rentals	12
5.4	Average Rental Duration	12
5.5	Top Rental Category	12
5.6	Rentals per Customer	12
5.7	Rental Rates by Film Category	12
5.8	Average Rental Duration by Film Category	13
5.9	Rentals by Film Title	13
5.10	Rental Distribution by Year	13
5.11	Monthly Rental Trend	13
5.12	Key Performance Indicators (KPIs)	13
6	Geographic Analysis	14
6.1	Total Revenue	14

6.2	Total Rentals	14
6.3	Total Customers	15
6.4	Average Revenue	15
6.5	Top Revenue-Generating Country	15
6.6	Top Revenue-Generating City	15
6.7	Revenue by City	15
6.8	Monthly Revenue Trend	15
6.9	Rentals by Country	15
6.10	Customers by Country	15
6.11	Revenue by Country (Map)	15
6.12	Key Performance Indicators (KPIs)	16
7	Film Analysis	16
7.1	Total Movies	16
7.2	Total Rentals	17
7.3	Average Rental Rate	17
7.4	Average Movie Length	17
7.5	Top Film	17
7.6	Top Category	17
7.7	Distribution of Films by Film Rating	18
7.8	Inventory Variation by Film Ratings	18
7.9	Film Information Table	18
7.10	Film Category Breakdown in Inventory	18
7.11	Revenue and Rental Rate by Film Category	18
7.12	Rentals Distribution by Films	18
7.13	Key Performance Indicators (KPIs)	18
8	Actor Analysis	19
8.1	Total Actors	19
8.2	Total Movies	19
8.3	Total Rentals	19
8.4	Average Films per Actor	20
8.5	Top Actor	20
8.6	Actor Revenue	20
8.7	Actor Details	20
8.8	Genre Preferences of Actors	20
8.9	Distribution of Films by Actors	21
8.10	Revenue Contribution by Actors	21
8.11	Key Performance Indicators (KPIs)	21
9	Dataset Limitations	21

9.1	Absence of Customer Rating Data	22
9.2	No Payment Method Information	22
9.3	Missing Employee Hire Date	22
9.4	Lack of Customer Demographic Information	22
9.5	Limited Language Diversity	22
9.6	No Direct Customer Satisfaction Metrics	22
9.7	No Store Distance or Location Coordinates	22
9.8	Limited Temporal Information	23
10	Overall Business Insights	23
10.1	Strong Customer Engagement	23
10.2	Revenue Concentration in Key Markets	23
10.3	Consistent Rental Activity	23
10.4	Balanced Inventory Distribution	23
10.5	Standardized Pricing Strategy	24
10.6	Geographic Diversity of Customers	24
10.7	Actor Participation Supports Content Diversity	24
10.8	Dataset Limitations Affect Certain Analyses	24
11	Recommendations	24
11.1	Strengthen Customer Loyalty Programs	24
11.2	Focus on High-Performing Markets	24
11.3	Optimize Inventory Management	25
11.4	Monitor Rental Trends	25
11.5	Improve Data Collection	25
11.6	Expand Multilingual Film Availability	25
11.7	Utilize Geographic Insights	25
11.8	Enhance Dashboard Monitoring	25
11.9	Leverage Actor and Category Popularity	25
11.10	Adopt Data-Driven Decision Making	26
12	Conclusion	26

List of Figures

1	Entity Relationship Diagram (ERD) and Power BI Data Model – Sakila DVD Rental Store	6
2	Customer Analysis Dashboard – Sakila DVD Rental Store	8
3	Revenue Analysis Dashboard – Sakila DVD Rental Store	10
4	Rental Analysis Dashboard – Sakila DVD Rental Store	12
5	Geographic Analysis Dashboard – Sakila DVD Rental Store	14
6	Film Analysis Dashboard – Sakila DVD Rental Store	17
7	Actor Analysis Dashboard – Sakila DVD Rental Store	20

1 Introduction

In today's data-driven business environment, organizations rely on analytics to gain valuable insights, improve operational efficiency, and make informed strategic decisions. Business Intelligence (BI) tools such as SQL and Power BI enable organizations to transform raw transactional data into meaningful information that supports decision-making and enhances overall business performance.

This project focuses on the analysis of the ****Sakila DVD Rental Store**** database, a widely used sample dataset that simulates the operations of a movie rental business. The dataset contains information related to customers, films, rentals, payments, inventory, actors, stores, staff, and geographic locations, providing an excellent foundation for performing comprehensive business analysis.

The primary objective of this project is to explore the dataset using SQL queries and visualize key business metrics through interactive Power BI dashboards. Various aspects of the business, including customer behavior, revenue generation, rental trends, film inventory, geographic performance, and actor participation, have been analyzed to identify meaningful patterns and actionable insights.

An interactive version of the Power BI dashboard has been published online and can be accessed using the following link:

<https://drive.google.com/file/d/1heKJDvkSsufTk3fQRPfgwM6r8z4WdCy2/view?usp=sharing>

The dashboard provides an integrated view of customer analytics, revenue performance, rental trends, geographic distribution, film analysis, and actor insights, enabling users to explore the business from multiple perspectives.

Although the Sakila dataset provides rich transactional information, certain limitations such as the absence of customer ratings, payment methods, employee hire dates, demographic details, and multilingual film records restrict some advanced analyses. These limitations have been identified and documented as part of the project.

Overall, this project demonstrates the practical application of SQL and Power BI in solving real-world business problems. The insights generated through this analysis can assist organizations in improving customer engagement, optimizing inventory management, enhancing revenue strategies, and making informed operational decisions based on data.

2 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) represents the logical structure of the Sakila DVD Rental database and illustrates how different entities are connected through primary

and foreign key relationships. It serves as the foundation for building the Power BI data model and enables integrated analysis across multiple business domains.

The data model consists of several interconnected tables, including Customer, Payment, Rental, Film, Inventory, Category, Actor, Staff, Store, Address, City, Country, Language, Film Category, Film Actor, and Film Text. These relationships allow seamless integration of transactional and master data for comprehensive reporting and analytics.

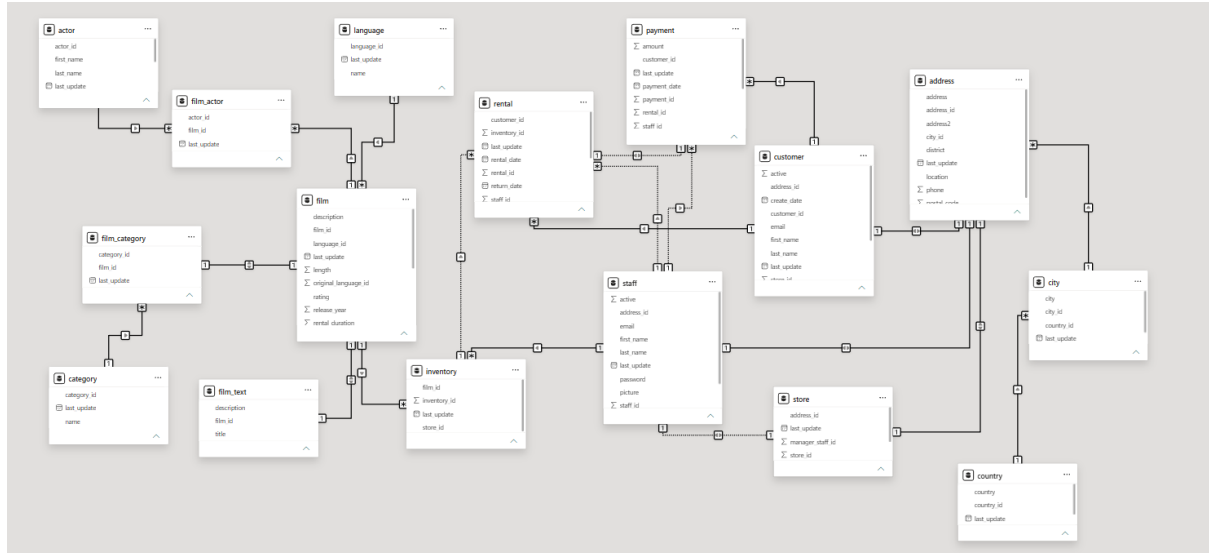


Figure 1: Entity Relationship Diagram (ERD) and Power BI Data Model – Sakila DVD Rental Store

2.1 Key Relationships

- **Customer → Rental:** Each customer can perform multiple rental transactions.
- **Customer → Payment:** Every payment is associated with a specific customer.
- **Rental → Inventory:** Each rental transaction is linked to a particular inventory item.
- **Inventory → Film:** Every inventory record corresponds to a specific film.
- **Film → Language:** Each film is associated with a language through the language id field.
- **Film → Category:** Films are connected to categories using the film category bridge table.
- **Film → Actor:** Movies are linked with actors through the film actor relationship table.
- **Store → Customer:** Customers are assigned to a particular store.

- **Store → Staff:** Each store is managed and operated by staff members.
- **Address → City → Country:** Geographic hierarchy is maintained through linked address, city, and country tables.

2.2 Importance of the Data Model

The relational structure enables efficient analysis of:

- Customer behavior and segmentation
- Revenue and payment trends
- Rental activity and inventory utilization
- Film and category performance
- Actor participation and popularity
- Geographic distribution of customers and revenue
- Store and staff performance

The ERD ensures data consistency and supports accurate cross-table analysis in Power BI by maintaining well-defined relationships between entities.

3 Customer Analysis

The Customer Analysis dashboard was designed to understand customer behavior, rental activity, revenue contribution, and geographic distribution. Various Power BI visualizations were used to identify valuable customers, monitor customer engagement, and analyze business performance from a customer perspective.

The dashboard analyzes the number of customers associated with different film ratings (G, PG, PG-13, R, and NC-17) to understand customer preferences across movie rating categories.

3.1 Active vs. Inactive Customers

A pie chart compares active and inactive customers. The analysis shows that the majority of customers are active, indicating strong customer engagement and retention.

3.2 Top Customers by Revenue

A bar chart identifies the highest revenue-generating customers based on their total spending. This helps the business recognize high-value customers and develop targeted loyalty programs.

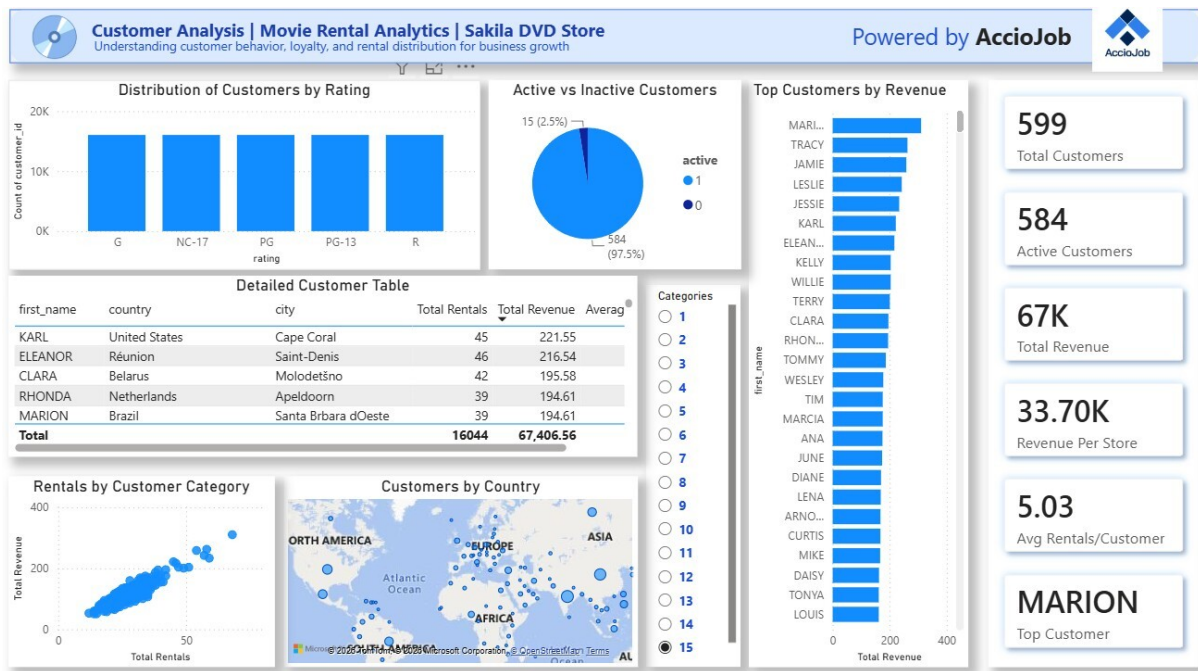


Figure 2: Customer Analysis Dashboard – Sakila DVD Rental Store

3.3 Detailed Customer Information

A detailed customer table displays customer name, country, city, total rentals, and total revenue, providing a comprehensive view of individual customer performance.

3.4 Rentals vs. Revenue Analysis

A scatter plot illustrates the relationship between total rentals and total revenue, showing that customers with more rentals generally contribute higher revenue.

3.5 Geographic Distribution of Customers

A map visualization displays customer locations across different countries, helping identify regions with higher customer concentration and potential business opportunities.

3.6 Key Performance Indicators (KPIs)

The dashboard includes several important business metrics:

- Total Customers: 599
- Active Customers: 584
- Total Revenue: 67K
- Revenue Per Store: 33.70K

- Average Rentals per Customer: 5.03
- Top Customer: MARION

Key Insights

- The business has a strong active customer base, with more than 97% of customers currently active.
- Revenue is largely driven by customers who make frequent rentals.
- High-value customers contribute significantly to total business revenue and can be targeted through personalized promotions and loyalty programs.
- Customer locations are spread across multiple countries, demonstrating a geographically diverse customer base.
- The positive relationship between rental frequency and revenue suggests that increasing customer engagement can directly improve business performance.

4 Revenue Analysis

The Revenue Analysis dashboard was developed to evaluate the financial performance of the Sakila DVD Rental Store. It provides insights into total revenue, monthly revenue trends, average customer payments, top-performing locations, category-wise revenue distribution, and geographic revenue patterns.

4.1 Total Revenue

The dashboard displays the overall revenue generated from all rental transactions. This KPI provides a quick summary of the business's total earnings.

4.2 Monthly Revenue Trend

A line chart illustrates how revenue changes over time, enabling the identification of peak and low-performing months.

4.3 Average Payment

The average payment amount per rental transaction is calculated to understand customer spending behavior and pricing consistency.

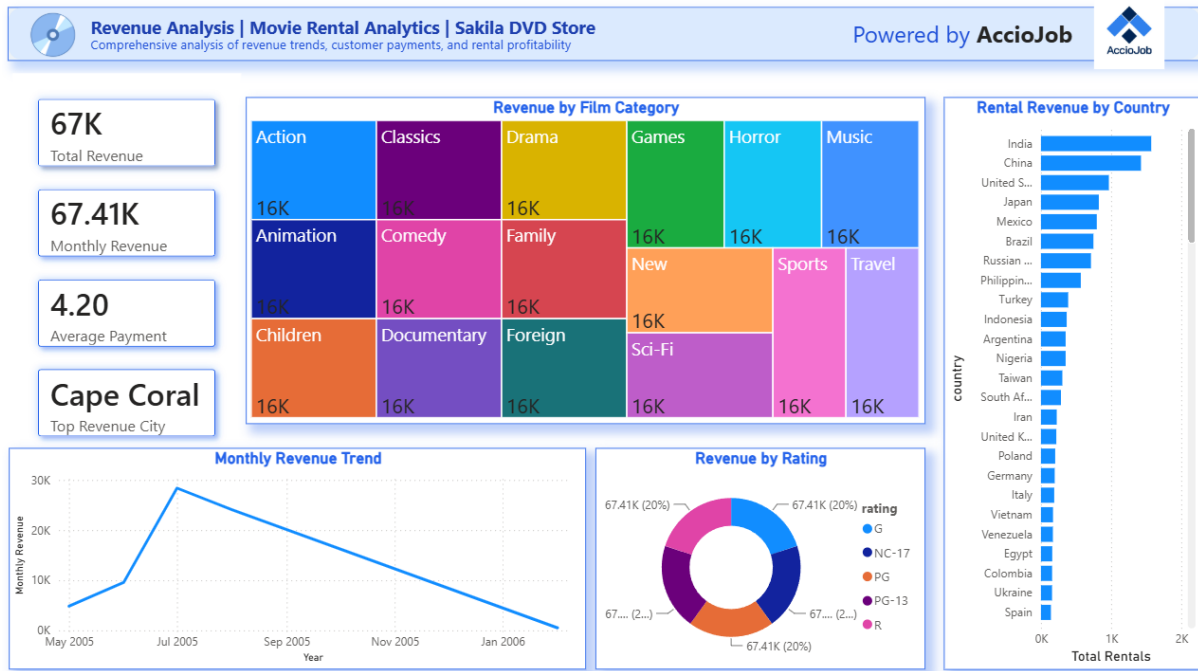


Figure 3: Revenue Analysis Dashboard – Sakila DVD Rental Store

4.4 Top Revenue-Generating City

The dashboard identifies the city that contributes the highest rental revenue, helping highlight key business markets.

4.5 Revenue by Film Category

A treemap visualizes revenue generated across different film categories, allowing comparison of category performance and identifying genres that contribute significantly to overall revenue.

4.6 Rental Revenue by Country

A horizontal bar chart compares rental revenue across different countries, providing insights into the geographical distribution of business income.

4.7 Revenue Distribution by Film Rating

A donut chart analyzes revenue generated from films with different content ratings (G, PG, PG-13, R, and NC-17), helping understand customer preferences based on movie ratings.

4.8 Key Performance Indicators (KPIs)

- Total Revenue: 67K

- Monthly Revenue: 67.41K
- Average Payment: 4.20
- Top Revenue City: Cape Coral

Key Insights

- The Sakila DVD Rental Store generated approximately 67K in total rental revenue.
- Revenue peaked during the middle of the observed period before gradually declining in later months.
- The average payment per rental transaction is approximately 4.20, indicating relatively consistent pricing across rentals.
- Cape Coral emerged as the highest revenue-generating city. Revenue is distributed across multiple film categories, with no single category overwhelmingly dominating total earnings.
- Geographic analysis shows that countries such as India and China contribute significantly to rental revenue compared to many other regions.
- Revenue generated from different film ratings appears relatively balanced, suggesting that customers rent movies across various content ratings.
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5 Rental Analysis

The Rental Analysis dashboard provides a comprehensive view of rental activity within the Sakila DVD Rental Store. It focuses on rental volume, return status, rental duration, category-wise performance, and monthly rental trends to evaluate customer rental behavior and operational efficiency.

5.1 Total Rentals

The dashboard displays the total number of rental transactions processed by the store, providing an overall measure of business activity.

5.2 Returned Rentals

The total number of successfully returned rentals is tracked to monitor rental completion and return performance.

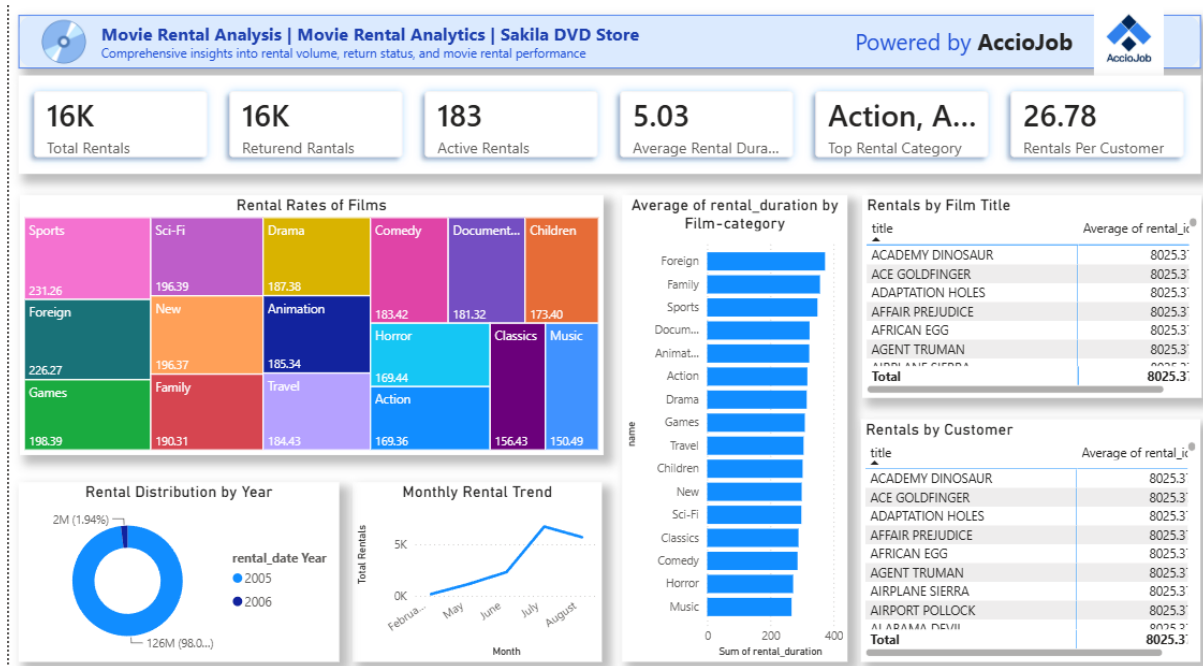


Figure 4: Rental Analysis Dashboard – Sakila DVD Rental Store

5.3 Active Rentals

The dashboard identifies rentals that are currently active or have not yet been returned, helping monitor ongoing transactions.

5.4 Average Rental Duration

The average number of days for which movies are rented is calculated to understand customer rental patterns.

5.5 Top Rental Category

The category with the highest rental activity is highlighted, indicating the most popular movie genre among customers.

5.6 Rentals per Customer

This KPI measures the average number of rentals made by each customer and helps evaluate customer engagement.

5.7 Rental Rates by Film Category

A treemap visualizes rental performance across different film categories, making it easy to compare category popularity.

5.8 Average Rental Duration by Film Category

A bar chart compares the average rental duration across different film categories, helping identify variations in rental behavior.

5.9 Rentals by Film Title

A detailed table presents rental information for individual film titles, allowing analysis at the movie level.

5.10 Rental Distribution by Year

A donut chart shows the proportion of rentals across different years, providing a historical overview of rental activity.

5.11 Monthly Rental Trend

A line chart illustrates rental trends over time, highlighting periods of increased or decreased customer demand.

5.12 Key Performance Indicators (KPIs)

- Total Rentals: 16K
- Returned Rentals: 16K
- Active Rentals: 183
- Average Rental Duration: 5.03 Days
- Top Rental Category: Action (based on dashboard output)
- Rentals per Customer: 26.78

Key Insights

- The Sakila DVD Rental Store has processed approximately 16K rental transactions, demonstrating consistent customer activity.
- Most rentals have been successfully returned, indicating an efficient rental management process.
- The average rental duration is around 5 days, suggesting standardized rental periods across customers.
- Certain film categories generate significantly higher rental activity than others, reflecting customer genre preferences.

- Monthly rental trends reveal fluctuations in customer demand, which can support inventory planning and promotional strategies.
- The analysis of rental duration by category provides insights into customer behavior and category-specific rental patterns.

6 Geographic Analysis

The Geographic Analysis dashboard provides insights into the regional performance of the Sakila DVD Rental Store by analyzing revenue, rentals, and customer distribution across different countries and cities. It helps identify high-performing locations and supports location-based business decision-making.

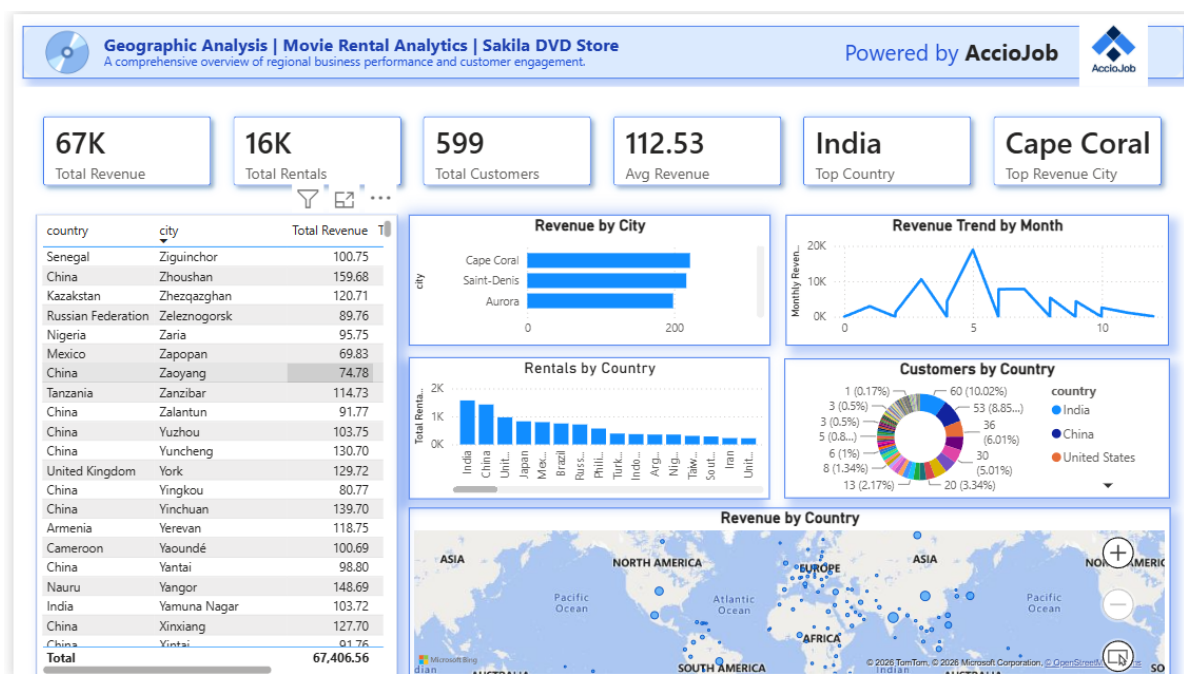


Figure 5: Geographic Analysis Dashboard – Sakila DVD Rental Store

6.1 Total Revenue

The dashboard displays the overall rental revenue generated across all geographic locations, providing a quick overview of business performance.

6.2 Total Rentals

The total number of rental transactions is presented to measure customer activity across different regions.

6.3 Total Customers

The dashboard shows the total number of registered customers, helping evaluate the size of the customer base.

6.4 Average Revenue

The average revenue metric provides an overview of the average earnings generated per business unit or transaction.

6.5 Top Revenue-Generating Country

The dashboard identifies the country that contributes the highest rental revenue, allowing management to recognize its strongest market.

6.6 Top Revenue-Generating City

The highest-performing city is highlighted to identify the most profitable local market.

6.7 Revenue by City

A bar chart compares rental revenue across different cities, making it easy to identify cities with the strongest business performance.

6.8 Monthly Revenue Trend

A line chart visualizes changes in revenue over time, helping detect seasonal patterns and business fluctuations.

6.9 Rentals by Country

A column chart compares rental activity across different countries and identifies regions with the highest customer demand.

6.10 Customers by Country

A donut chart illustrates the distribution of customers among various countries, providing insights into market coverage.

6.11 Revenue by Country (Map)

A geographic map displays revenue distribution across the world, enabling visual identification of high-revenue regions and supporting location-based strategic planning.

6.12 Key Performance Indicators (KPIs)

- Total Revenue: 67K
- Total Rentals: 16K
- Total Customers: 599
- Average Revenue: 112.53
- Top Country: India
- Top Revenue City: Cape Coral

Key Insights

- The Sakila DVD Rental Store generated approximately 67K in total revenue from customers across multiple countries.
- India is the highest revenue-generating country, making it one of the most important markets for the business. Cape Coral contributes the highest city-level revenue among all locations.
- Rental activity is distributed across many countries, although a few regions account for a larger share of total rentals.
- Customer distribution demonstrates a globally diverse customer base, indicating international market presence.
- Monthly revenue trends reveal fluctuations over time, suggesting possible seasonal changes in rental demand.
- The geographic map provides a clear visualization of revenue concentration and highlights opportunities for targeted regional marketing.

7 Film Analysis

The Film Analysis dashboard provides a comprehensive overview of the movie inventory and rental performance in the Sakila DVD Rental Store. It focuses on film ratings, rental demand, inventory distribution, category performance, movie characteristics, and revenue generation to support data-driven business decisions.

7.1 Total Movies

The dashboard displays the total number of movies available in the inventory, providing an overview of the size of the film catalog.

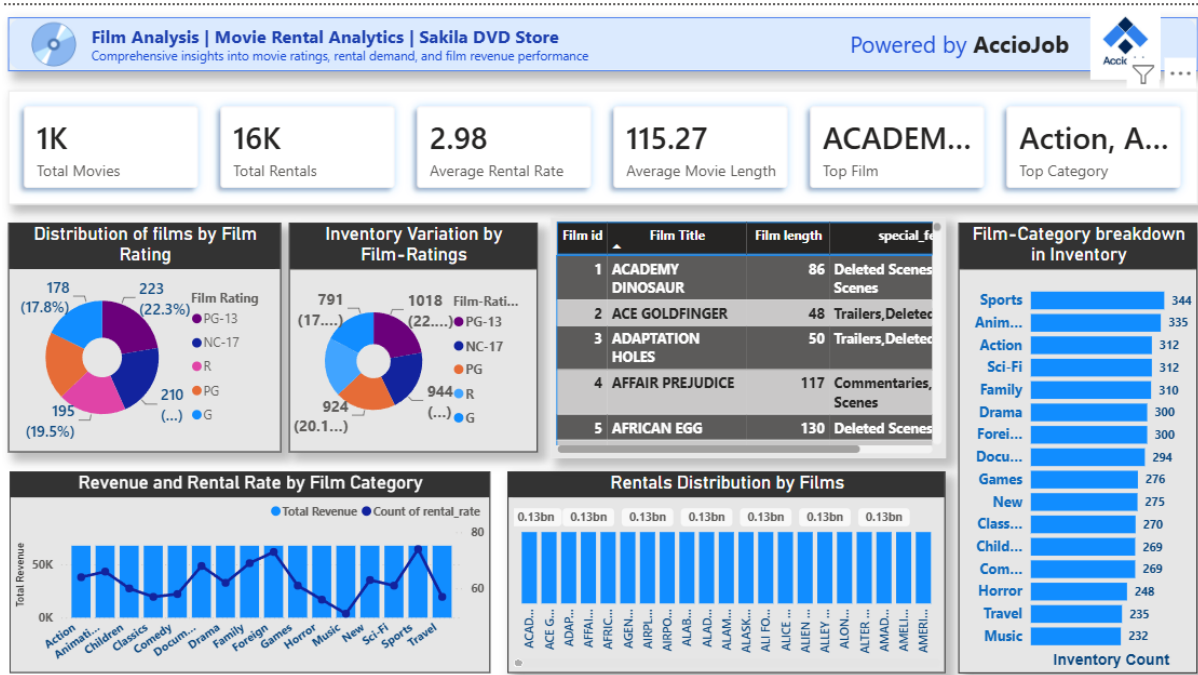


Figure 6: Film Analysis Dashboard – Sakila DVD Rental Store

7.2 Total Rentals

The total number of rental transactions is presented to measure overall customer demand for movies.

7.3 Average Rental Rate

The average rental rate of all films is calculated to evaluate the pricing strategy adopted by the rental store.

7.4 Average Movie Length

The dashboard shows the average duration of movies in minutes, helping understand the typical length of films available in the inventory.

7.5 Top Film

The most popular or highest-performing film is highlighted based on rental activity.

7.6 Top Category

The dashboard identifies the film category with the highest rental demand and customer engagement.

7.7 Distribution of Films by Film Rating

A donut chart visualizes the distribution of films across different content ratings (G, PG, PG-13, R, and NC-17), providing insights into the composition of the movie catalog.

7.8 Inventory Variation by Film Ratings

Another donut chart compares inventory counts across different film ratings, helping evaluate stock allocation for each rating category.

7.9 Film Information Table

A detailed table presents film-level information, including film title, film length, and available special features, allowing users to analyze individual movies.

7.10 Film Category Breakdown in Inventory

A horizontal bar chart displays the number of inventory copies available for each film category, helping identify categories with larger or smaller inventory levels.

7.11 Revenue and Rental Rate by Film Category

A combined chart compares total revenue and rental rate across different film categories, providing insights into category performance and pricing effectiveness.

7.12 Rentals Distribution by Films

A column chart illustrates rental distribution among individual films, enabling comparison of rental activity across titles.

7.13 Key Performance Indicators (KPIs)

- Total Movies: 1K
- Total Rentals: 16K
- Average Rental Rate: 2.98
- Average Movie Length: 115.27 Minutes
- Top Film: Academy Dinosaur
- Top Category: Action (based on dashboard output)

Key Insights

- The Sakila DVD Rental Store maintains an inventory of approximately 1,000 movies, providing customers with a diverse selection of films.
- The average rental rate is 2.98, indicating a consistent pricing strategy across the movie catalog.
- The average movie length is approximately 115 minutes, suggesting that most films are of standard feature-film duration.
- Film inventory is distributed across multiple categories, with some categories containing significantly more copies than others.
- Revenue and rental activity vary across film categories, highlighting differences in customer preferences and category performance.
- The analysis of film ratings shows a balanced distribution of movies among different content classifications.
- Category-wise inventory and rental trends can support better stock management and promotional planning.

8 Actor Analysis

The Actor Analysis dashboard provides insights into actor performance, movie participation, rental activity, and revenue contribution within the Sakila DVD Rental Store. It helps evaluate the popularity of actors, their involvement in films, and their indirect impact on rental demand and business revenue.

8.1 Total Actors

The dashboard displays the total number of actors available in the database, providing an overview of the talent represented in the movie catalog.

8.2 Total Movies

The total number of movies associated with the actors is presented to understand the overall size of the film library.

8.3 Total Rentals

The dashboard reports the total number of rental transactions, reflecting customer engagement with movies featuring different actors.

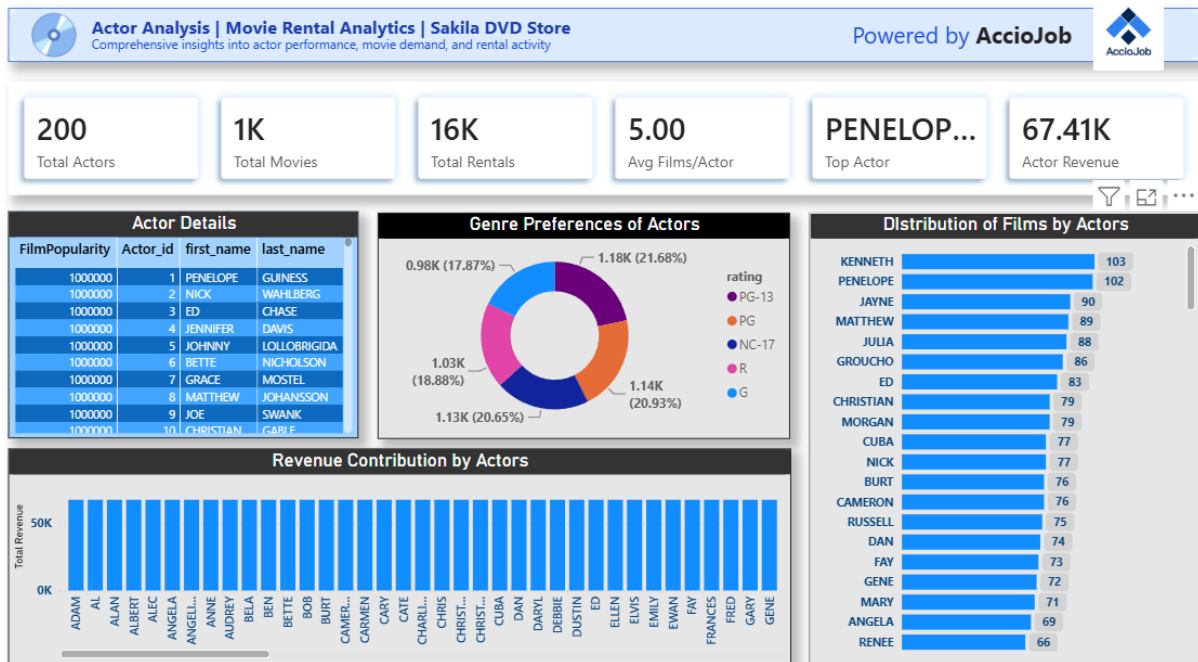


Figure 7: Actor Analysis Dashboard – Sakila DVD Rental Store

8.4 Average Films per Actor

This KPI calculates the average number of films associated with each actor, helping evaluate actor participation across the movie catalog.

8.5 Top Actor

The actor with the highest movie participation or popularity is highlighted, identifying the most prominent performer in the dataset.

8.6 Actor Revenue

The dashboard displays the overall revenue generated from rentals involving movies featuring the listed actors.

8.7 Actor Details

A detailed table provides actor-specific information, including Actor ID, first name, last name, and popularity metrics for individual analysis.

8.8 Genre Preferences of Actors

A donut chart illustrates the distribution of actors across films with different content ratings (G, PG, PG-13, R, and NC-17), providing insights into the types of movies in which actors most frequently appear.

8.9 Distribution of Films by Actors

A horizontal bar chart compares the number of films associated with each actor, helping identify actors with the largest film portfolios.

8.10 Revenue Contribution by Actors

A column chart visualizes the revenue generated by movies associated with different actors, supporting analysis of actor-related business performance.

8.11 Key Performance Indicators (KPIs)

- Total Actors: 200
- Total Movies: 1K
- Total Rentals: 16K
- Average Films per Actor: 5.00
- Top Actor: Penelope Guinness
- Actor Revenue: 67.41K

Key Insights

- The Sakila dataset contains 200 actors participating across approximately 1,000 films.
- On average, each actor appears in 5 movies, indicating a balanced distribution of actor participation.
- Penelope Guinness is identified as the top actor based on the dashboard metrics.
- The distribution of films varies among actors, with some actors appearing in significantly more movies than others.
- Revenue generated from actor-associated films contributes substantially to the overall business performance.
- The analysis of film ratings demonstrates that actors participate across multiple content classifications, reflecting diversity in movie genres and target audiences.

9 Dataset Limitations

Although the Sakila DVD Rental dataset provides comprehensive information related to customers, films, rentals, payments, inventory, staff, and stores, it has several limitations that restrict certain business analyses.

9.1 Absence of Customer Rating Data

The dataset does not contain customer ratings, reviews, or feedback scores. Therefore, it is not possible to evaluate customer satisfaction or identify locations with the highest and lowest customer ratings.

9.2 No Payment Method Information

The ‘payment’ table records transaction amounts but does not include the payment method (e.g., Cash, Credit Card, Debit Card, or Online Payment). As a result, the distribution of sales by payment method cannot be analyzed.

9.3 Missing Employee Hire Date

The ‘staff’ table does not contain employee hiring or joining dates. Consequently, staff employment duration and workforce tenure analysis cannot be performed.

9.4 Lack of Customer Demographic Information

The dataset does not include customer age, date of birth, gender, occupation, or income details. Therefore, analyses based on age groups or demographic segmentation are not possible.

9.5 Limited Language Diversity

Although the ‘language’ table contains multiple languages such as English, Italian, Japanese, Mandarin, French, and German, all 1,000 films in the dataset are available only in English. This limits multilingual and language-based preference analysis.

9.6 No Direct Customer Satisfaction Metrics

There are no fields related to customer experience, service quality, complaints, or satisfaction surveys, making customer experience analysis impossible.

9.7 No Store Distance or Location Coordinates

The dataset does not provide geographical coordinates or distance information between customers and stores. Therefore, proximity-based rental behavior cannot be accurately analyzed.

9.8 Limited Temporal Information

While transaction dates are available, the dataset lacks detailed historical business events or promotional information that could explain seasonal variations in rentals or revenue.

10 Overall Business Insights

The analysis of the Sakila DVD Rental Store dataset provides valuable insights into customer behavior, revenue generation, rental trends, inventory management, geographic performance, film characteristics, and actor participation. By integrating SQL analysis with interactive Power BI dashboards, meaningful business patterns and opportunities for improvement have been identified.

10.1 Strong Customer Engagement

The business maintains a highly active customer base, with the majority of registered customers actively participating in rental transactions. Loyal and regular customers contribute significantly to overall revenue, highlighting the importance of customer retention strategies.

10.2 Revenue Concentration in Key Markets

Revenue analysis indicates that certain countries and cities contribute a larger share of total business income. Markets such as India and cities like Cape Coral generate substantial revenue and should remain priority locations for business growth and marketing initiatives.

10.3 Consistent Rental Activity

The store has processed more than 16,000 rental transactions, demonstrating stable customer demand. Rental trends also reveal seasonal fluctuations, which can be used to improve inventory planning and promotional campaigns.

10.4 Balanced Inventory Distribution

The movie inventory is distributed across multiple categories and film ratings, ensuring a diverse catalog for customers. Categories such as Sports, Animation, and Action maintain relatively high inventory levels to meet customer demand.

10.5 Standardized Pricing Strategy

The average rental rate remains relatively consistent across the movie catalog, indicating a stable pricing model that simplifies rental decisions for customers while maintaining predictable revenue generation.

10.6 Geographic Diversity of Customers

Customers are distributed across numerous countries and cities, demonstrating the international reach of the Sakila DVD Rental Store. Geographic analysis enables region-specific business strategies and targeted marketing efforts.

10.7 Actor Participation Supports Content Diversity

The database includes 200 actors participating in approximately 1,000 films. Actor analysis provides additional insights into film portfolios and supports content recommendation strategies.

10.8 Dataset Limitations Affect Certain Analyses

Several business questions could not be fully answered due to missing information in the dataset, including customer ratings, payment methods, employee hire dates, customer demographics, and multilingual film records. Future data collection should address these gaps to enable more advanced analytics.

11 Recommendations

Based on the SQL analysis and Power BI dashboards, the following recommendations are proposed to improve business performance and support data-driven decision-making.

11.1 Strengthen Customer Loyalty Programs

Regular and loyal customers contribute a significant portion of the overall revenue. The business should introduce reward programs, membership benefits, and personalized offers to encourage repeat rentals and improve customer retention.

11.2 Focus on High-Performing Markets

Countries and cities generating the highest revenue, such as India and Cape Coral, should receive greater marketing attention and business investment. Expanding promotional campaigns in these regions can further increase profitability.

11.3 Optimize Inventory Management

Film categories with consistently high rental demand should maintain sufficient inventory levels to avoid stock shortages. At the same time, low-demand categories should be reviewed periodically to optimize inventory costs.

11.4 Monitor Rental Trends

Monthly rental and revenue trends should be continuously monitored to identify seasonal demand patterns. This information can be used to schedule promotional campaigns and improve inventory planning during peak rental periods.

11.5 Improve Data Collection

The current dataset lacks important information such as customer ratings, payment methods, employee hire dates, and customer demographic details. Capturing these attributes would enable more comprehensive business analysis and support better strategic decisions.

11.6 Expand Multilingual Film Availability

Although multiple languages exist in the language table, all available films are in English. Adding movies in different languages would increase content diversity and allow analysis of language-based customer preferences.

11.7 Utilize Geographic Insights

Geographic analysis should be used to identify high-performing and underperforming regions. Region-specific marketing strategies and localized promotional campaigns can help improve customer acquisition and revenue generation.

11.8 Enhance Dashboard Monitoring

The Power BI dashboards should be updated regularly with new transactional data to provide real-time insights into customer behavior, rental performance, revenue trends, and inventory status.

11.9 Leverage Actor and Category Popularity

Popular actors and high-performing film categories can be highlighted in promotional campaigns and recommendation systems to increase customer engagement and rental activity.

11.10 Adopt Data-Driven Decision Making

Management should use the insights generated from SQL queries and Power BI dashboards to support strategic planning, optimize business operations, and improve overall customer satisfaction through evidence-based decision-making.

12 Conclusion

The Sakila DVD Rental Analytics project successfully demonstrates how SQL and Power BI can be used together to transform raw transactional data into meaningful business insights. By analyzing customer behavior, revenue trends, rental performance, inventory distribution, film characteristics, geographic patterns, and actor participation, the project provides a comprehensive understanding of the overall business operations.

The interactive Power BI dashboards effectively visualize key performance indicators (KPIs) and enable users to explore important metrics such as total revenue, rental activity, customer distribution, inventory levels, and regional performance. These visualizations support faster and more informed business decision-making.

The analysis also highlights several important findings, including the significant contribution of loyal customers to revenue, the concentration of business performance in specific countries and cities, and the need for effective inventory management based on rental demand. Furthermore, the project identifies limitations within the Sakila dataset, such as the absence of customer ratings, payment method information, employee hire dates, demographic details, and multilingual film records, which restrict certain advanced analyses.

Overall, this project illustrates the practical application of business intelligence techniques using SQL and Power BI. The insights generated can help organizations optimize operations, improve customer engagement, enhance inventory planning, and make data-driven strategic decisions. With additional business attributes and real-world data, the analytical capabilities of this solution can be further expanded to provide even deeper and more valuable insights. efficiency.